

HARSH SHARMA

AIML Engineer

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PROFESSIONAL SUMMARY

Result-driven final-year B-Tech student with hands-on experience in Machine Learning, Deep Learning, Large Language Models, Retrieval-Augmented Generation (RAG) systems and deep learning pipelines for production use. Skilled in **Python, SQL, LangChain, LangGraph, Pytorch and vector database** delivering end-to-end ML solutions including image captioning (CNN-LSTM), medical image classification and agentic LLM workflows. Passionate about building scalable AI systems and eager to apply this foundation to production ready ML systems in a full time-role.

CORE TECHNICAL SKILLS

Programming Languages: Python, Java, SQL

Machine Learning: Supervised & Unsupervised Learning, Statistical Modeling, Transfer Learning, Model Fine-tuning, Feature Engineering, Model Evaluation

Deep Learning: CNNs, LSTMs, Transformers, RNNs, Encoder-Decoder Architectures

Generative AI & LLMs: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Multi-Agent Systems, OpenAI GPT, Embeddings, Vector Databases, Lora and Qlora

NLP: NLTK, Text Preprocessing, Tokenization, Semantic Search

Frameworks & Libraries: PyTorch, scikit-learn, NumPy, Pandas, Matplotlib, Langchain, LangGraph

Data & Databases: MySQL, MongoDB, Pinecone, ChromaDb

Tools & Platforms: Docker, Git, GitHub, Jupyter Notebook

CS Fundamentals: Data Structures & Algorithms, OOPs, DBMS

PROFESSIONAL EXPERIENCE

Data Scientist Intern | Zidio Development

08/2025 – 10/2025

Remote

- Designed and built an end-to-end **image captioning pipeline using a CNN-LSTM encoder-decoder architecture** trained on the MS COCO dataset (83,000+ images), delivering real-time inference through a Gradio web interface.
- Engineered a **transfer learning strategy** by fine-tuning a pretrained ResNet-50 encoder while training the LSTM decoder from scratch, reducing training time by 40% without compromising caption quality.
- Deployed the model as an interactive web application**, enabling real-time image upload and caption generation, and collaborated with cross-functional teams to deliver a demo-ready system within the 2 month internship.

KEY PROJECTS

Medical Chatbot — LLM-Based RAG System

Tech Stack: Python, LangChain, Pinecone, Streamlit, OpenAI

GitHub: <https://github.com/HARSH-0000/medical-chatbot>

- Built an end-to-end **Retrieval-Augmented Generation (RAG) chatbot** combining a Large Language Model with a Pinecone vector store to deliver context-aware responses on medicines and drug interactions.
- Designed **prompt engineering strategies** to ground LLM outputs in a structured medical encyclopedia knowledge base, improving response accuracy and reducing hallucinations.
- Exposed the system through a **Streamlit web interface** for real-time user queries.

Neuroscan- Brain Tumor Classification

Tech Stack: Python, PyTorch, VGG16, Transfer Learning, Streamlit

GitHub: https://github.com/HARSH-0000/brain_tumour

- Trained a **CNN-based deep learning classifier** on brain MRI scans across four categories (glioma, meningioma, pituitary, no tumor), achieving 95%+ test accuracy through transfer learning on a pretrained VGG16 model.
- Deployed the model as a **web application** enabling users to upload MRI scans and receive real-time tumor classification with confidence scores.

Multi-Agent Research System

Tech Stack: Python, LangChain, OpenAI, Tavily, BeautifulSoup

GitHub: https://github.com/HARSH-0000/Multi_agent_system

- Architected a **4-stage agentic workflow (Search → Reader → Writer → Critic)** using LangChain tool-based agents with Tavily search and BeautifulSoup scraping for automated information retrieval, synthesis, and evaluation at scale with a latency of less than 10 sec.

EDUCATION

B-Tech, Artificial Intelligence & Machine Learning

2023 – 2027

GH Raison College of Engineering and Management, Pune | **CGPA: 8.6 / 10**

Relevant Coursework: Machine Learning, Deep Learning, Artificial Intelligence, Gen AI, Data Structures & Algorithms, DBMS

CERTIFICATIONS & ACHIEVEMENTS

Generative AI - IBM | **Cloud Computing** - Infosys | **Introduction to Data Science** - Cisco | **Claude 101** - Anthropic

Languages: English (Professional), Hindi (Native), German (Elementary)